EXAM: FINAL PRESENTATION

The final presentation should be a technical presentation to showcase the use case defined, agreed with the professor and developed throughout the course. It can be anything related to the topics of the course (modelling, simulation, digital twin) and could include a (live) demo if there is the opportunity to produce simulation results. *Tip: Imagine you are presenting your technical product or simulation tool to investors and want to demonstrate how it works and what are its benefits.*

Presentation time should be between 10 and 15 minutes and it will be followed by approximately 10 minutes of questions, not only on the topic of the presentation but also on all the topics covered throughout the course (in particular complementary to the topic of the presentation).

CRITERIA FOR EXAM PRESENTATION

- **Technical understanding of the course topics**

Via the exam presentation, students should demonstrate their knowledge of the topics covered throughout the course and should be able to describe them in detail wherever applicable to the use case selected. Additional questions may follow to complement the topic of the presentation.

- **Technical depth and/or breadth**

Depending on the use case selected, students should either demonstrate breadth and understanding of the big picture (e.g. multi-disciplinary understanding for end-to-end digital twin use cases) or in-depth knowledge of one specific modelling or simulation aspect (e.g. with a software demo – a simulation that has been set up by the students or a code that has been written and/or run by the students). Showing both breadth and depth would be a further plus.

- **Creativity**

Via the selection of the use case, the definition of the scope of the problem and the approach to solving the problem, students should show critical thinking and creativity. The professor provides feedback throughout the course for the selection of the use case, in order to guide the thinking. If starting from existing literature or software, students are encouraged to make modifications to the problem in order to add their creative contribution.

- **Literature review**

Students should be able to search through literature for the state-of-the-art technologies and data available to solve the problem selected and should show ability to study, understand and explain a problem from the literature, as well as creatively modify it if needed. References to the literature should be included in the presentation (a final 'references' slide is sufficient).

- **Presentation skills**

Students should demonstrate the ability to clearly describe a technical topic to an audience of non-experts, providing sufficient technical details to put the use case into context and describing the problem definition, the approach and the potential solution (or solutions). The presentation should be readable, understandable and have no typos or formatting issues. Time should be managed appropriately by the students and the presentation should be delivered in its entirety within the allocated time. Some time will be allocated to questions afterwards. Students should answer the questions in a clear, concise but as detailed as possible manner.